How large is the equality-of-factors automaton? Empirical evidence and construction strategy for an open problem of Khodier

Andrew Hingston

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Abstract

For a k-automatic sequence T produced by an m-state DFAO, the *equality of factors* predicate $FE(i,j,l) := \forall t\,(t < l \Rightarrow T[i+t] = T[j+t])$ defines a language whose minimal automaton size $|FE|$ is of interest both computationally (Walnut and related tools routinely blow up while constructing it) and structurally: Khodier's 2026 thesis states as Open Problem 1 that no class of examples with *exponential* blowup in m is known, and conjectures that one exists. We report a large empirical sweep (random k-uniform morphisms, $k \in \{2,3\}$, $m \leq 7$) that finds no such family in the bulk ($|FE|$ scales like 2–4 m^3 at the median, with a small unresolved tail), identify that most apparent construction-time blowups are digit-order artefacts rather than genuine largeness of the minimal automaton, isolate "lossy coding of a group automaton" as the one lead that empirically inflates $|FE|$, and give a reverse-first-determinization construction strategy that solves several inputs on which a current build of Walnut fails at equal memory. On the theory side we report an unconditional, referee-checked result: for the generalised Thue–Morse family G_p (a p-state DFAO), $|FE_{G_p}|_{\mathrm{msd}} = \Theta(p^3)$, exactly $p^3 + 8$ for $3 \leq p \leq 24$ – the first infinite family with $|FE|$ pinned to a matching polynomial order. We also report a conditional upper bound $|FE| \leq m^4 + m^6 + m^8\Lambda(T)^2$ in terms of a computable, morphism-only invariant Λ, and state honestly what remains open: no exponential family, and no unconditional upper bound smaller than the literature's 2^{9m^2}, has been found.

Contents

1 Introduction and problem statement

Automatic sequences – infinite sequences produced by finite automata reading the base-k digits of n – support a decision procedure (implemented in tools such as Walnut) that reduces first-order statements about the sequence to automaton constructions. A recurring primitive in that procedure is the *equality of factors* predicate

$$FE(i,j,l) \;:=\; \forall t\,\big(t < l \Rightarrow T[i+t] = T[j+t]\big),$$

asserting that the length-l factors of T starting at positions i and j agree. FE underlies statements about repetitions, palindromes, critical exponents, and recurrence, and its automaton is frequently built as an intermediate step in larger queries even when it is not the final object of interest.

Mazen Khodier's 2026 PhD thesis (*New Methods for Analyzing the Properties of Automatic Sequences*, University of Waterloo, Chapter 8) records that the size of the minimal automaton for FE, relative to the size m of the DFAO producing T, is open. Verbatim:

Open Problem (Khodier 2026, Chapter 8)**.** *Give a characterization for the k-automatic sequences that have an exponential blowup in size when constructing the "equality of factors" automaton. Currently no such class of examples is known. This issue has two distinct aspects. The first concerns the size of the minimal automaton for the equality of factors predicate relative to the size of the morphism. We believe that this relationship is exponential, but currently, no class of examples is known that proves this behavior. The second aspect involves determining the most efficient strategy for constructing this automaton […]*

The thesis records one existing upper bound, for a different but logically equivalent formulation

$$FE_2(i,j,n) := \forall u,v.\ (u \ge i \,\wedge\, u < i+n \,\wedge\, u+j = v+i) \Rightarrow T[u] = T[v],$$

namely $|FE_2| \le 2^{9m^2}$, and a worked anecdote: the Tribonacci word's minimal FE has only 26 states, but the naive one-shot Walnut construction of it peaked at 323,831,403 intermediate states and 300 GB before finishing. That anecdote is the seed of the present study: it shows the two aspects of the open problem are decoupled – the *final* automaton can be small even when the *construction* is enormous – and raises the question of how general that decoupling is.

This paper reports on an empirical and partly theoretical investigation of both aspects, using a purpose-built Rust automaton engine (§2) with a Brzozowski reverse-first construction strategy and hard memory guards, run at scale over random and structured families of automatic sequences. §2 describes the engine and sweep design. §3 reports the random-ensemble table, the digit-order artefact finding, structured-family results, and the coding-spectrum lead. §4 gives the construction strategy and a like-for-like benchmark against Walnut. §5 states what is unconditionally proved and what remains conjectural, following an independent adversarial referee pass (`paper/proof-verdict.md`). §6 states the polynomial conjecture with an honest confidence level. §7 and §8 close with limitations and reproducibility.

On overclaiming. Every number in this paper traces to a file under `results/*.json`, a table in `docs/TARGET1.md` or `docs/TARGET1-TABLE.md`, or a lemma/theorem in `paper/proof-*.md` whose status has been independently checked (`paper/proof-verdict.md`). Section 5 includes *only* what the referee verdict marks PROVED; everything the verdict marks fitted, heuristic, or partially machine-verified is reported as such, not as a theorem.

2 Method

2.1 Engine

The automaton engine (`engine/`, Rust) is driven by a small command language over stdin: `mode msd|lsd` selects most- or least-significant-digit-first reading; `def T k m start w_0..w_{m-1} coding` defines a k-uniform, m-state DFAO with an output coding; `let NAME(args) formula` defines a named predicate in a first-order language over $\forall/\exists$, indexing $T[i+t]$, and boolean connectives `& | => !=`, with `$NAME(...)` for predicate composition; `? formula` decides a closed formula; `mem` reports live/peak allocator bytes. $|FE|$ throughout is the number of states of the minimal *complete* DFA, including the dead state (Walnut's convention differs by exactly 1; this is corrected for in all Walnut comparisons in §4).

2.2 Guard

A single earlier sweep run crashed the host machine by launching many unbounded engine processes in parallel. Three independent guards now prevent this (`docs/GUARD.md`): (1) a hard allocator budget inside the engine (`AM_MEM_MB`, default 2048 MB; on exceeding it the process exits cleanly with status 3 and no partial state); (2) `explore/engine.py`, the sole sanctioned way to launch the engine from Python, providing admission control (a job does not start until free RAM exceeds a floor and kernel memory pressure is normal), a watchdog that kills over-budget children, and guaranteed cleanup on exit; (3) a system-level LaunchAgent that kills any engine process exceeding an absolute RSS ceiling regardless of how it was launched. A `FAIL budget` outcome in a sweep is a datum – the automaton genuinely exceeded the ceiling – not an artefact of the harness.

2.3 Sweep design

The primary random sweep (`explore/blowup.py`, plus rescue passes) draws k-uniform admissible morphisms with a binary coding from a sampler named PRISM, for $k \in \{2,3\}$ and DFAO size $m \in \{2,\ldots,7\}$, targeting up to 40 admissible sequences per (k,m) cell. For each sequence, $|FE|$ is measured as the minimal DFA size of $FE(i,j,l)$. A sequence that fails (memory budget or timeout) in msd is retried in lsd, and vice versa; sequences that fail in both digit orders are further retried under Khodier's alternative FE_2 formulation and under a forced small-subset-cap ladder

(§4). The final merged table (`docs/TARGET1-TABLE.md`, built by `explore/final_table.py` from `results/blowup.json`, `results/blowup_retry.json`, `results/blowup_residue.json`, `results/blowup_residu` `results/blowup_residue3.json`) keeps one number per sequence, preferring an msd value (msd and lsd sizes for the same predicate genuinely differ) and falling back to an lsd-only or still-censored classification when no msd value was obtained in any pass.

Beyond the random ensemble, structured families are swept by dedicated scripts (`explore/thin_families*.py` `struct_families.py`, `gtm_full.py`, `blowup_features.py`) covering thin/counting sets (fixed number of 1-bits), pattern-containment sets, digit-sum-mod-p sequences under both the faithful group coding and a lossy binary coding, and a feature-correlation study relating prefix statistics (subword complexity growth, run length, right-special count) to $|FE|$ within a size class.

3 Results

3.1 Random ensemble: no exponential family in the bulk

Table 1 reproduces the final uncensored per-(k, m) quartile table (`docs/TARGET1-TABLE.md`), merged over five rescue passes. Of 412 dispatched sequences, 356 resolved to an msd value, 45 resolved only in lsd (not directly comparable, hence excluded from the quartiles), and 11 remain still-censored after five rescue passes – all in the two hardest cells, $k = 2, m = 6$ (one sequence) and $k = 3, m = 7$ (ten sequences).

Table 1: Random-ensemble $|FE|$ (msd) by (k, m), quartiles over resolved sequences. n = dispatched, cens = still-censored. Source: `docs/TARGET1-TABLE.md`, `results/final_table.json`.

k	m	n	cens	min	q1	med	q3	max	med/m^3	$\log_2(\max)/m$
2	2	8	0	3	6	8	10	15	0.94	1.95
2	3	33	0	9	34	51	96	218	1.89	2.59
2	4	24	0	24	58	161	404	549	2.52	2.28
2	5	39	0	97	260	340	403	924	2.72	1.97
2	6	40	1	201	474	608	966	2124	2.81	1.84
2	7	40	0	200	685	882	1134	1770	2.57	1.54
3	2	28	0	3	7	17	22	38	2.12	2.62
3	3	40	0	10	50	72	100	190	2.69	2.52
3	4	40	0	135	216	250	348	532	3.91	2.26
3	5	40	0	112	328	424	509	1604	3.40	2.13
3	6	40	0	311	540	752	872	2356	3.48	1.87
3	7	40	10	521	722	847	1085	3480	2.47	1.68

Power-law fits to the medians beat exponential fits by roughly an order of magnitude in log-residual sum of squares (`docs/TARGET1-TABLE.md`):

$$k = 2: \quad \text{med} \approx 0.666\, m^{3.81}\ (\text{SSE} = 0.18) \qquad \text{vs.} \qquad \text{med} \approx 2.44 \cdot 2.496^m\ (\text{SSE} = 1.26),$$
$$k = 3: \quad \text{med} \approx 2.11\, m^{3.23}\ (\text{SSE} = 0.21) \qquad \text{vs.} \qquad \text{med} \approx 6.42 \cdot 2.168^m\ (\text{SSE} = 1.07).$$

$\log_2(\max)/m$, the quantity that would grow without bound under a true exponential family, instead *falls* with m in five of six columns per k. The largest msd value found anywhere in the ensemble is $|FE| = 3480$ ($k = 3, m = 7$, `def T 3 7 0 044 513 202 604 520 421 061 1111110`), surfaced only via the forced small-cap ladder of §4 – first-pass msd and every other formulation failed on it.

This is evidence about the *typical* case under one random-morphism sampler, not a proof about all automatic sequences; a single adversarial family would settle Open Problem 1(A) regardless of what typical sequences look like. The 11 still-censored sequences, concentrated entirely in $k = 3, m = 7$, are exactly where such a family would have to live and remain unresolved.

3.2 Finding: most apparent blowups are digit-order artefacts

Of 47 sequences that failed the memory budget in msd during the first sweep pass, 45 (96%) finished in lsd at modest size (64–989 states for $k = 3$, $m \in \{3, 4\}$). Every $k = 3$, $m \in \{3, 4\}$ failure was such an artefact: the *construction* blew up in one digit order while the *minimal* automaton was small and reachable in the other. This reproduces, at scale across hundreds of sequences, exactly the pattern of Khodier's Tribonacci anecdote (26 final states, 323 million intermediate states) – suggesting the decoupling between construction cost and final size is common, not a one-off. It also means aspect (B) of Open Problem 1 (construction strategy) is a real and separate problem from aspect (A) (minimal size): most of what looks like evidence for exponential blowup in a naive single-pass sweep is not evidence for aspect (A) at all.

3.3 Structured families

Table 2 summarizes the structured-family sweeps (`docs/TARGET1.md`, "Structured families"). All values are minimal $|FE|$, msd unless stated; a dash marks an engine failure at the stated budget.

Table 2: Structured families, $|FE|$ vs. parameter. Source: `docs/TARGET1.md`.

Family	Mode	Growth
exactly c ones, base 2 ($m = c + 2$)	msd	52, 183, –, 1042, 2008, 3463 ($\sim c^3$)
exactly c ones, base 2	lsd	41, 242, 1133, 4227, 13461, 38504 (ratios $5.9, 4.7, 3.7, 3.2, 2.9$)
at most c ones, base 2	msd	51, 108, –, 327, 502, 730 ($\sim c^{2.5}$)
contains 1^r in binary ($m = r + 1$)	msd	7, 91, –, –, 998, 1713, 2700, 4004 (ratios ~ 1.5)
contains 10^{r-1}	msd	13, 54, –, –, 587, 1026, 1651 (polynomial)
$s_2(n)$ mod p, full output ($m = p$)	msd	15, 35, –, 133, 224, 351, 520, 737 ($\sim p^{2.5}$)
$s_2(n)$ mod p, full output	lsd	22, 39, 56, 77, 102, 131, 164, 201 (exactly quadratic)
[$s_2(n)$ mod $p \neq 0$], binary code	msd	15, 190, 698, 1877, –, –, –, – (ratios $12.7, 3.7, 2.7$)

Reading (`docs/TARGET1.md`): thin/counting sets and pattern-containment sets both give polynomial growth. The generalised-Thue–Morse family with *full* output is polynomial and exactly quadratic in lsd; the naive heuristic that FE forces $i \equiv j \pmod{2^p}$ (hence 2^p msd states) is refuted by direct measurement and retracted in that document – the congruence is forced globally along the whole window, so the msd automaton never needs a sliding window. The one family that grows fast within the range measured is [$s_2(n)$ mod $p \neq 0$], the **lossy binary coding** of the same p-state group automaton, discussed next.

3.4 The coding-spectrum lead

Collapsing $s_2(n)$ mod p from a faithful p-letter output to the single bit [$s_2(n) \not\equiv 0$ mod p] keeps the DFAO at p states but multiplies $|FE|$ by roughly $100\times$ at $p = 4$ in lsd (6154 vs. 56; `docs/TARGET1.md`). This is the one effect in the entire sweep where a size-preserving change to the sequence (recoding the output alphabet, not changing the transition structure) produces a large multiplicative jump in $|FE|$, and it is the empirical seed of the theoretical work in §5: the relevant invariant is not "big automaton" but "lossy coding of a group automaton," where the FE automaton must track a pair of group elements rather than their difference.

3.5 The $3p^4$ fit, and its retraction

An earlier fit to the singleton-coding family $T_p = [s_2(n) \equiv 1 \bmod p]$ reported $|FE_{T_p}| \approx 3p^4$ to within 2.5% using six points, $p = 3, \ldots, 8$: $190, 698, 1877, 3971, 7243, 11988$. This fit is **not** carried forward as established. The referee review (`paper/proof-verdict.md`, on `proof-lower.md` §5) and the family document itself (`proof-family.md` §5.4) both flag it as unsafe: at $p = 3$, $3p^4 = 243$ against a measured 190, a 28% error, not 2.5%; and the local log-log slope of the same six points *falls through* 4 (from 4.52 to 3.77 over $p = 3..8$), while a cubic fit $(44.4p^3 - 205.4p^2 + 308.5p - 88.7)$ matches the same data with residual ≤ 5.7. The decisive point that would separate a cubic from a quartic law, $p = 9$ msd, does not fit in a 6 GB memory guard and remains unresolved (confirmed again during the referee pass: allocator budget exceeded, return code 3). The status of T_p's growth exponent is therefore **open**, not $\Theta(p^4)$ as earlier stated.

4 Construction strategy and the Walnut benchmark

4.1 The ladder

Across every hard case encountered, a small forward-subset-construction cap (e.g. 50,000 subsets) that fails fast into a Brzozowski reverse-first determinization consistently beats a large forward cap by orders of magnitude: an exact-$c = 3$ counting sequence resolves to 482 states in effectively no time under the small-cap-then-reverse ladder, versus exhausting a 6 GB budget under a large forward cap; $[s_2 \equiv a \bmod 5]$ resolves at 30 MB versus blowing 3.3 GB (`docs/TARGET1.md`). This ladder – small forward cap, Brzozowski reverse, larger forward cap, Brzozowski reverse again – is now the engine's default construction strategy.

4.2 Like-for-like benchmark against Walnut

`bench/README.md` reports a controlled comparison against Walnut 8-dev (head 2026-06-15, OpenJDK 26, `java -Xmx6g`), both run with a 6 GB memory ceiling and, for the engine, a 15-minute wall-clock timeout on the Walnut side. Both tools count minimal DFA states; the engine's count includes the dead state, Walnut's does not, so `ours = Walnut + 1` whenever both finish – itself an independent cross-check of every $|FE|$ number reported elsewhere in this paper.

On easy inputs the two tools are equivalent – both instant, same automaton. On the FE-hard inputs (group automata with lossy codings, PRISM-1, the sweep tail), Walnut 8-dev at 6 GB fails on 7 of 9 and needs 7 minutes on an eighth, while the engine answers 8 of 9 in 0–170 s at the same memory. The difference, per `bench/README.md`, is construction strategy – reverse-first determinization with a small forward cap, and the lsd/msd switch – not raw speed; Walnut has a `reverse` command and could adopt the same strategy, but it is not its default.

Fairness caveats, quoted verbatim from `bench/README.md`. "Walnut got its default forward msd construction with no expert reformulation (Khodier's self-verifying predicates would rescue some of these); a bigger heap would rescue some more; and this is Walnut on the predicate its own authors document as the pathological one. It is a benchmark of one predicate family, not of the tools."

Table 3: $|FE|$ and wall time, engine vs. Walnut 8-dev, 6 GB each. Source: `bench/README.md`.

Sequence	Ours (states)	s	Walnut (states)	s
Thue–Morse	15	0.0	14	0.3
period-doubling	8	0.0	7	0.2
Rudin–Shapiro	68	0.1	67	0.4
paperfolding	44	0.0	43	0.2
Cantor	17	0.0	16	0.2
Mephisto	14	0.0	13	0.3
PRISM-a	24	0.0	23	0.3
PRISM-d	82	0.5	81	0.9
champion-$m5$	199	0.0	198	6.1
$k3m3$-artefact-b	71	0.0	70	18.0
$k3m3$-artefact-a	216	0.0	215	444.5
PRISM-1 (k=4, m=6)	467	38.6	OOM	251.5
$[s_2 \equiv a \bmod 3]$	190	1.5	timeout	900
$[s_2 \equiv a \bmod 4]$	698	0.2	OOM	313
$[s_2 \equiv a \bmod 5]$	1877	2.1	OOM	92
$[s_2 \equiv a \bmod 6]$	3971	20.9	OOM	53
tail-a (k=2, m=7)	1165	139	OOM	77
tail-b (k=3, m=5)	1000	168	OOM	119
tail-c (k=2, m=6)	fail (535)	–	OOM	164

4.3 `learnfe`: guess-and-verify construction

Table 3's one remaining direct-construction failure, tail-c (k=2, m=6, `def T 2 6 0 05 23 44 42 51 10 000010`), which stalled at 535 states under a 6 GB budget, is resolved by a second construction strategy implemented as the engine command `learnfe` (`docs/LEARNFE.md`). Rather than compiling the universally-quantified formula $FE(i,j,l) = \forall t\,(t < l \Rightarrow T[i+t] = T[j+t])$ directly – the projection over t that causes the blowup – `learnfe` builds a candidate DFA by Kearns–Vazirani active learning against a membership oracle $FE(i,j,l) \Leftrightarrow l \le \mathrm{LCP}(i,j)$, then *verifies* the candidate against a recurrence

$$H(i,j,l) \iff \big(l = 0 \text{ or } (T[i]{=}T[j] \text{ and } H(i{+}1, j{+}1, l{-}1))\big)$$

that only FE satisfies (a one-line induction on l; `docs/LEARNFE.md` §2), so the reported automaton is exactly FE regardless of how the candidate was reached – there is no notion of an approximately-correct answer. On tail-c itself, `learnfe` completes at **1382** states in 14.2 s and 50 MB peak. On the 17 cases where both constructions finish, they agree exactly, checked inside the engine as `A i,j,l. $FE(i,j,l) <=> $G(i,j,l)` → `TRUE` (`docs/LEARNFE.md` §6.1), and independently, without any automaton machinery, by a pure-Python brute force over morphism-generated prefixes diffing every $FE(i,j,l)$, $i,j,l < B$ against the learned automaton (416/416 for Rudin–Shapiro at B=12 in both digit orders, 826/826 for tail-c at B=14, 325/325 for tail-b at B=11).

Because tail-c now has a verified value, the Walnut comparison of Table 3 becomes **9/9**: on all nine FE-hard inputs (the four group-automaton lossy codings, PRISM-1, and the four sweep-tail sequences), the engine now produces a verified answer at 6 GB where Walnut 8-dev fails on seven and needs seven minutes on an eighth.

`learnfe` also resolves three further sequences from the still-censored tail of §3 that neither digit order of the direct construction could build at a 2.5 GB budget in 360 s (`results/learnfe_censored.json`): tail-b (k=3, m=5) at 1000 states, 25.7 s, 179 MB; `c-3.7d` (k=3, m=7) at 1147 states, 44.4 s, 109 MB; and `c-3.7h` (k=3, m=7) at 781 states, 23.2 s, 80 MB. Nine of the thirteen sequences attempted in that pass remain unresolved at this budget – all $k = 3$, where the 27-letter track alphabet makes each equivalence query an order of magnitude more expensive. `learnfe` is not uniformly faster than the direct construction: on easy inputs (e.g. Thue–Morse) the direct construction wins by orders of magnitude, since `learnfe`'s cost tracks the *answer* size rather than any intermediate blowup. It is a second tool for aspect (B) of Open Problem 1, not a replacement for the ladder of §4.

5 Towards a proof

This section reports only what an independent adversarial referee pass (`paper/proof-verdict.md`, checking every load-bearing lemma by hand and, where finite, by an independent brute force not written by the original authors) marked **PROVED**. Results marked as fits, heuristics, or partially machine-verified are labelled accordingly; none are stated as theorems.

5.1 Proved: a matching-order family, $|FE_{G_p}| = \Theta(p^3)$

Let $G_p[n] := s(n) \bmod p$ (the digit sum of n in base 2, reduced mod p), the fixed point of the 2-uniform morphism $a \mapsto a\,(a{+}1)$ on $\mathbb{Z}_p$ with the identity coding: a p-state DFAO ($m = p$), the *generalised Thue–Morse word*.

Theorem 5.1 (Referee-verified; `proof-family.md` Thm. 4.4, 4.5)**.** *For every $p \geq 3$, the minimal msd DFA of FE_{G_p} has at least p^3 states, and*

$$p^3 \;\leq\; |FE_{G_p}|_{\text{msd}} \;\leq\; 34p^3 + 98p^2 + 41p \;=\; O(m^3), \qquad |FE_{G_p}|_{\text{lsd}} \;\leq\; 128p^2 + 160p + 19 \;=\; O(m^2).$$

The lower bound rests on a parity-rigidity lemma (a factor of G_p of length ≥ 4 determines the parity of its occurrence positions) and a Myhill–Nerode argument distinguishing p^3 residuals by an explicit construction of prefixes; the upper bound is by inspection of an explicit finite state space of size $O(p^3)$ (resp. $O(p^2)$ in lsd). Both directions were independently re-derived by hand and, where finite, checked by brute force ($p = 2..7$, up to 2,058,000 triples for the characterisation lemma) by the referee pass, with 0 mismatches.

Separately, and only for each individual p in the range checked (a machine-verified statement, not a proof for all p):

$$|FE_{G_p}|_{\text{msd}} = p^3 + 8, \qquad |FE_{G_p}|_{\text{lsd}} = 2p^2 + 3p + 12, \qquad 3 \leq p \leq 24$$

(exception $p = 2$, ordinary Thue–Morse: 15 and 22). Engine values reproduced by the referee at $p = 3, 7, 11, 13, 16$: msd $= 35, 351, 1339, 2205, 4104$; lsd $= 39, 131, 287, 389, 572$, matching the closed forms exactly.

This is, to our knowledge, the first infinite DFAO family for which $|FE|$ is pinned to a matching polynomial order – evidence *against* exponential blowup for this specific class, against a literature ceiling of 2^{9m^2} (Khodier 2026) for the general case.

5.2 Proved (conditional): a computable predictor Λ, and $|FE| \leq m^4 + m^6 + m^8\Lambda(T)^2$

Theorem 5.2 (Referee-verified; `proof-upper.md` Thm. 5.4, and supporting Lemma 3.1, Cor. 3.2, Thm. 4.3)**.** *For any k-automatic T with minimal msd DFAO of size m, the Myhill–Nerode residual* *of an FE prefix (I, J, L) with $L \geq 2$ is determined by exactly* 8 *DFAO states plus one further datum* Θ *(the middle language), which depends on (I, J, L) only through two subsets of Q^3. Consequently there is a morphism-only invariant $\Lambda(T)$, computable in time polynomial in m^3 and Λ without ever constructing FE, such that*

$$|FE_{\mathrm{msd}}(T)| \ \leq \ m^4 + m^6 + m^8\,\Lambda(T)^2.$$

This was checked by `paper/proof-upper-check.py` with 0 violations on the authors' worked examples, and independently re-run by the referee on five adversarial DFAOs the authors had not tested (an $m = 4$ extremal automaton, a thin set, a ruler family, and both codings of $\mathbb{Z}_4$), again 0 violations. Table 4 gives measured $(\Lambda, |FE|)$ pairs.

Table 4: Λ vs. measured $|FE|$ on adversarial examples. Source: `paper/proof-verdict.md` §1.

Example	Λ	$\#\Theta$	$\lvert FE\rvert$ (engine)
C_4 champion (`0 01 22 33 10 / 1110`)	74	61	1152
ruler $r = 4$	35	61	113
exactly-2-ones	30	10	183
$\mathbb{Z}_4$ singleton coding	28	24	698
$\mathbb{Z}_4$ faithful coding	4	2	72

What this theorem does and does not give. Λ separates the extremes (4 for the faithful group automaton, 74 for the densest $m = 4$ example found), and it is empirically the only invariant in this study that separates a faithful coding from a lossy coding of the *same* underlying automaton. But Λ is *not* monotone in $|FE|$ across families (the ruler has $\Lambda = 35$, $|FE| = 113$; the singleton $\mathbb{Z}_4$ coding has $\Lambda = 28$, $|FE| = 698$ – larger $|FE|$, smaller Λ), and nothing in the theorem bounds Λ itself. Since $\Lambda \leq 2^{\min(\gamma, m^3)}$ for a related quantity γ that is not otherwise controlled, the unconditional content of Theorem 5.2 is $|FE| = 2^{O(m^3)}$ – *weaker* than the literature's 2^{9m^2} (Khodier 2026), not an improvement on it. The referee pass is explicit that Theorem 5.4, taken as a bound, is a reparametrisation of the obstruction, not a solution; its real value is Λ as a cheap, engine-free predictor computed without ever building FE.

5.3 Not proved: the coding-lead is not closed

An earlier draft of `proof-family.md` claimed that a level decomposition of the lossy-coding mechanism (Prop. 5.3: at bit-level $\kappa = v_2(j - i)$ the effective coding is $\chi_{\min(\kappa, p-1)}$, saturating at the identity coding by level $p - 2$) "closes" the lossy-coding lead as a route to exponential blowup. The referee pass identifies this as **circular**: level 0 of the grading *is* the original singleton-coded predicate FE_{T_p} restricted to odd differences, so the argument's conclusion – "$|FE_{T_p}|$ is not exponential unless some single level is already exponential" – reduces to "$|FE_{T_p}|$ is not exponential unless $|FE_{T_p}|$ is exponential." There is also an unaddressed possibility that the p levels combine multiplicatively rather than additively (the msd automaton cannot in general read off $\kappa = v_2(j-i)$ from a bounded prefix), which is exactly the shape an exponential family would take; this does not happen in the data measured so far, but that is a measurement, not a proof. No upper bound of any kind for $|FE_{T_p}|$ exists in any of the three source documents. The correct summary, per the referee verdict, is: the level decomposition is a correct structural fact about FE_{G_p}'s coding, and it is *not* a proof that lossy-coded group automata cannot be an exponential family.

5.4 Gaps, stated explicitly

- **Open Problem 1(A) is not answered by this paper.** No family with provably exponential $|FE|$ relative to m is exhibited, in either direction. Theorem 5.1 is evidence *against* exponential blowup for one natural family; it is not a resolution of the general question.

- **No upper bound smaller than 2^{9m^2} is proved.** Theorem 5.2's unconditional content is $2^{O(m^3)}$, strictly weaker than the literature bound it was hoped to improve on.

- $|FE_{T_p}|$**'s growth exponent is open**, not 4 as an earlier fit suggested (§3); the decisive data point ($p = 9$, msd) exceeds the 6 GB memory guard.

- **The closed forms $p^3 + 8$ and $2p^2 + 3p + 12$ are machine-verified for $3 \leq p \leq 24$, not proved for all p.** The construction underlying them is proved correct as a design from the exact characterisation (Cor. 3.5), but was brute-force cross-checked only to $p \leq 6$; values for $17 \leq p \leq 24$ rest on a single implementation.

- **The extremal-family claim in the random-tail direction is data-limited.** A separately measured extremal ladder-construction champion at $m = 5$ (`def T 2 5 0 01 22 33 44 10 11110`) resolves at $|FE| = 3846$ msd, beating the 1200-sequence sample champion (2415) by 1.6× and continuing the family beyond $m = 4$ (1152); but only three points (15, 1152, 3846 at $m = 3, 4, 5$) are available, their consecutive $\log_2$ increments are $4.14, 2.13, 1.74$ – falling, but slowly enough that this is not decisive evidence either for or against a bounded-exponent law, and $m = 6, 7$ for this specific family remain unmeasured.

6 Conjecture

Conjecture 6.1. *For every k-automatic sequence T produced by a minimal m-state DFAO, $|FE(T)|_{\mathrm{msd}} = O(m^4)$.*

Honest confidence: low-to-moderate, and explicitly not the position of Khodier's thesis, which states the opposite belief. In favour: every random-ensemble median through $m = 7$ (§3) is polynomial with exponent 3.2–3.8; every structured family measured (thin sets, pattern-containment, faithful group codings) is polynomial; the one proved infinite family (Theorem 5.1) is exactly cubic; and $\log_2(\max)/m$ falls, not rises, with m across the whole random ensemble. Against: the sweep only reaches $m \leq 7$ and 11 sequences remain permanently unresolved in exactly the cell ($k = 3, m = 7$) where an exponential family would have to live; the one family whose exponent is empirically *not* settled (T_p, singleton coding of a group automaton) is precisely the family the field's one lead points at, and its exponent could be 3, 4, or – unproved to be excluded – unbounded; and no unconditional polynomial upper bound exists for any family, only the $2^{O(m^3)}$ bound of Theorem 5.2, which is weaker than the known 2^{9m^2}. This conjecture should be read as "no evidence found for the exponential belief, in a large but bounded search," not as a claim that the belief is wrong.

7 Limitations

- The random ensemble is drawn from one sampler (PRISM) of admissible morphisms and reaches only $m \leq 7$; a single adversarial family well outside this distribution would settle Open Problem 1(A) regardless of the ensemble's shape.

- 11 sequences (all in $k = 3, m = 7$) remain still-censored after five rescue passes and several hours of machine time; they are the natural place to look next.
- The construction-strategy benchmark (§4) compares one predicate family (equality of factors) between two tools at one memory ceiling; it is explicitly stated in `bench/README.md` not to be a general claim of superiority over Walnut, which received no expert reformulation.
- The $3p^4$ fit for T_p was retracted after independent review found a 28% error at $p = 3$ and a falling local exponent; readers of earlier internal documents (`docs/TARGET1.md`) should treat that number as superseded by §3 of this paper.
- The Λ invariant (Theorem 5.2) is not itself known to be polynomial in m; its unconditional bound is weaker than the pre-existing literature bound, and its value here is as a cheap empirical predictor, not a proof technique for Open Problem 1(A) in either direction.

8 Reproducibility

All numbers in this paper trace to committed artefacts.

- **Engine:** `engine/` (Rust; `cargo build --release`), driven only via `explore/engine.py` (`engine.run` / `engine.pool`), never raw `subprocess` with multiple workers (`docs/GUARD.md`).
- **Random ensemble:** `explore/blowup.py` (first pass), `explore/blowup_retry.py` (lsd rescue), residue passes for the double-failure tail; merged by `explore/final_table.py` into `results/final_table.json`, tabulated in `docs/TARGET1-TABLE.md`.
- **Structured families:** `explore/thin_families.py`, `explore/thin_families2.py`, `explore/struct_fami` `explore/gtm_full.py`, `explore/blowup_features.py`.
- **Walnut benchmark:** `bench/walnut_fe.py`, results and fairness caveats in `bench/README.md`.
- **Theory:** `paper/proof-upper.md`, `paper/proof-family.md`, `paper/proof-lower.md`; machine checks in `paper/proof-upper-check.py` and `paper/verdict-checks/` (`fam_check.py`, `thm44.py`, `prop51.py`, `exh_nonbin.py`); the full adversarial referee pass is `paper/proof-verdict.md`, whose one-line verdict per document and per-lemma status table this section summarizes.
- **Guard infrastructure:** `docs/GUARD.md`, `explore/memguard.sh`.

To reproduce the referee's independent checks:

```
cd /Users/andrew/maths
.venv/bin/python paper/proof-upper-check.py
.venv/bin/python paper/verdict-checks/fam_check.py
.venv/bin/python paper/verdict-checks/thm44.py
.venv/bin/python paper/verdict-checks/prop51.py
.venv/bin/python paper/verdict-checks/exh_nonbin.py 3
.venv/bin/python paper/verdict-checks/exh_nonbin.py 4
```

Acknowledgements

This work is a direct response to Open Problem 1 of Mazen Khodier's 2026 thesis, and the Tribonacci construction anecdote reported there motivated the digit-order-artefact investigation in §3.

References

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